BMATE101 – MATHEMATICS-I (EEE)

MODULE–3 : ORDINARY DIFFERENTIAL EQUATIONS

FIRST-ORDER DIFFERENTIAL EQUATIONS

1. Differential Equation – Basic Idea

A **differential equation** is an equation involving:

- an independent variable
- a dependent variable
- one or more derivatives of the dependent variable

2. Order and Degree

- **Order**: Highest order of derivative present
- **Degree**: Power of the highest order derivative (when equation is polynomial in derivatives)

Example:

$$\left(\frac{d^2y}{dx^2}\right)^2 + \frac{dy}{dx} = x$$

Order = 2, Degree = 2

3. First-Order Linear Differential Equation

Standard Form

$$\boxed{\frac{dy}{dx} + P(x)y = Q(x)}$$

Solution Method

1. Identify

$$P(x),\, Q(x)$$

2. Find **Integrating Factor (IF)**:

$$\boxed{\text{IF} = e^{\int P(x)\,dx}}$$

3. Multiply the equation by IF
4. Solve:

$$\frac{d}{dx}(y \cdot \text{IF}) = Q(x) \cdot \text{IF}$$

Example 1

Solve:

$$\frac{dy}{dx} + y = e^x$$

Here,

$$\begin{aligned}
&P(x) = 1, Q(x) = e^x \\
&\text{IF} = e^{\int 1 dx} = e^x \\
&\frac{d}{dx}(ye^x) = e^{2x} \\
&ye^x = \int e^{2x} dx = \frac{e^{2x}}{2} + C
\end{aligned}$$

4. Bernoulli's Differential Equation

Standard Form

$$\boxed{\frac{dy}{dx} + P(x)y = Q(x)y^n, n \neq 0,1}$$

Method

1. Divide entire equation by y^n
2. Substitute:

$$v = y^{1-n}$$

3. Reduce to linear form and solve

Example 2

Solve:

$$\frac{dy}{dx} + y = y^2$$

Divide by y^2:

$$y^{-2}\frac{dy}{dx} + y^{-1} = 1$$

Let $v = y^{-1}$

$$\frac{dv}{dx} - v = -1$$

This is linear → solve using IF.

5. Exact Differential Equations

General Form

$$\boxed{M(x, y)\, dx + N(x, y)\, dy = 0}$$

Condition for Exactness

$$\boxed{\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}}$$

Solution Procedure

1. Integrate Mw.r.t x
2. Integrate Nw.r.t y
3. Combine results + constant

Example 3

$$(2xy + y^2)dx + (x^2 + 2xy)dy = 0$$
$$\frac{\partial M}{\partial y} = 2x + 2y \frac{\partial N}{\partial x} = 2x + 2y$$

Exact

6. Reducible to Exact Equation

If the equation is **not exact**, it can sometimes be made exact by multiplying by an **Integrating Factor** depending on:

- $x$alone
- or $y$alone

7. Orthogonal Trajectories

Definition

If two families of curves intersect at right angles, they are called **orthogonal trajectories**.

Procedure

1. Find differential equation of given family
2. Replace $\frac{dy}{dx}$by $-\frac{dx}{dy}$
3. Solve the new equation

8. Applications of First-Order Differential Equations

- Electrical circuits (L-R and C-R circuits)
- Growth and decay models
- Heat conduction
- Radioactive decay

CIRCUIT APPLICATIONS & NON-LINEAR DIFFERENTIAL EQUATIONS

9. L–R Circuit (Application of Differential Equations)

Consider an electrical circuit containing:

- Inductance L
- Resistance R
- Applied voltage E

Let $i(t)$be the current at time t.

Differential Equation of L–R Circuit

Using Kirchhoff's voltage law:

$$L\frac{di}{dt} + Ri = E$$

Rewriting:

$$\frac{di}{dt} + \frac{R}{L}i = \frac{E}{L}$$

This is a **first-order linear differential equation**.

Solution

Integrating factor:

$$\text{IF} = e^{\int \frac{R}{L}dt} = e^{\frac{R}{L}t}$$
$$i(t) = \frac{E}{R}\left(1 - e^{-\frac{R}{L}t}\right)$$

Interpretation

- Current gradually increases
- Final steady current $= \frac{E}{R}$

10. C–R Circuit

A C–R circuit consists of:

- Capacitance C
- Resistance R
- Applied voltage E

Let $q(t)$be the charge on the capacitor.

Differential Equation of C–R Circuit

$$R\frac{dq}{dt} + \frac{q}{C} = E$$

This is also a **first-order linear differential equation**.

Solution

$$q(t) = CE\left(1 - e^{-\frac{t}{RC}}\right)$$

Current:

$$i(t) = \frac{dq}{dt} = \frac{E}{R}e^{-\frac{t}{RC}}$$

Interpretation

- Charge increases exponentially
- Current decays with time

11. Non-Linear Differential Equations

A differential equation is **non-linear** if:

- The dependent variable or its derivatives appear with power ≠ 1
- Or multiplied together

Example:

$$\left(\frac{dy}{dx}\right)^2 + y = x$$

12. General and Singular Solutions

General Solution

- Contains an arbitrary constant
- Represents a family of curves

Example:

$$y = Cx$$

Singular Solution

- Obtained by eliminating the constant
- Represents a curve touching all curves of the family

13. Clairaut's Differential Equation

Standard Form

$$\boxed{y = px + f(p)}$$

where

$$p = \frac{dy}{dx}$$

General Solution

Replace $p$by constant C:

$$\boxed{y = Cx + f(C)}$$

Singular Solution

Differentiate w.r.t C:

$$x + f'(C) = 0$$

Eliminate Cbetween:

$$y = Cx + f(C)\text{and}x + f'(C) = 0$$

Example

$$y = px + p^2$$

General solution:

$$y = Cx + C^2$$

Singular solution:

$$x + 2C = 0 \Rightarrow C = -\frac{x}{2}$$

$$y = -\frac{x^2}{4}$$

14. Applications of Module–3

- Electrical circuits (L–R, C–R)
- Growth and decay problems
- Heat transfer
- Mechanical motion
- Signal modelling